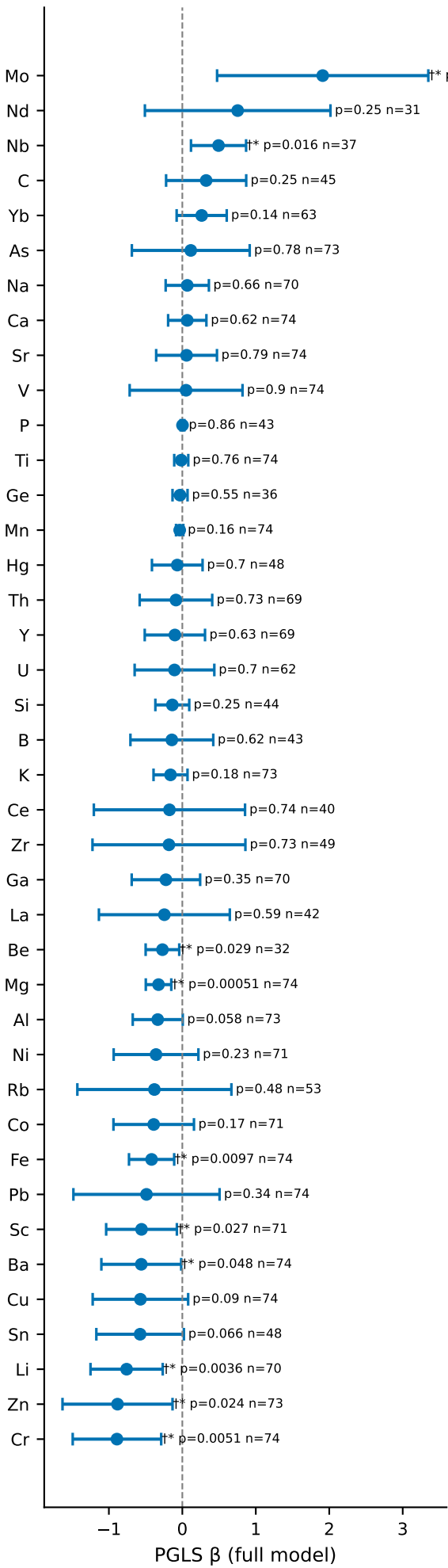


SPIRE MAGs × USGS: KO density → element log-ppm
(full controls: genome, Levins B, lat/lon, soil chemistry)

Cofactor KO density
(SPIRE MAGs, n≥15 genera)



Resistance KO density
(SPIRE MAGs, n≥15 genera)

